

REGION 4

Set up all six orders of integration for

$$\iiint_{R_4} f(x, y, z) dV,$$

where R_4 is the region bounded by

$$x + y + z = 1, \quad x = 0, \quad y = 0, \quad \text{and} \quad z = 0.$$

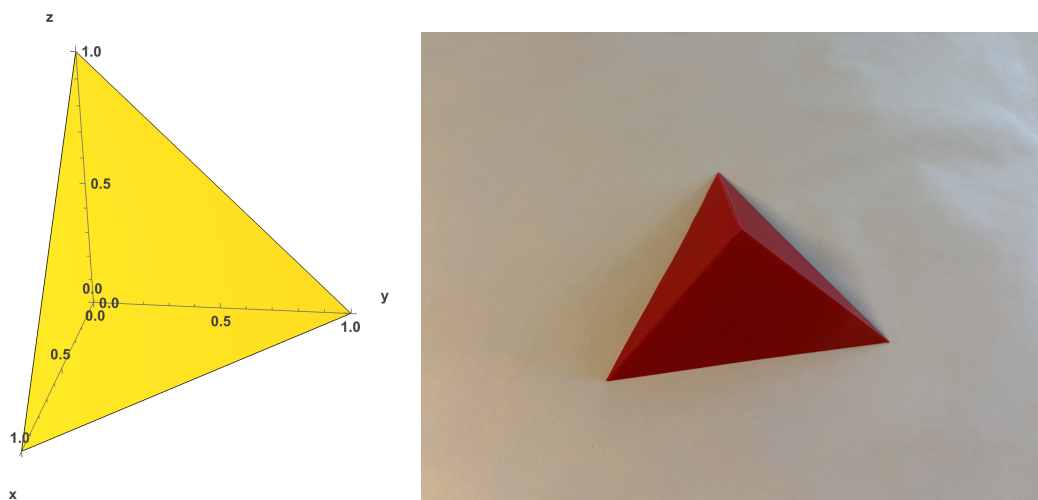


FIGURE 1. Computer-modeled and 3D-printed views of R_4

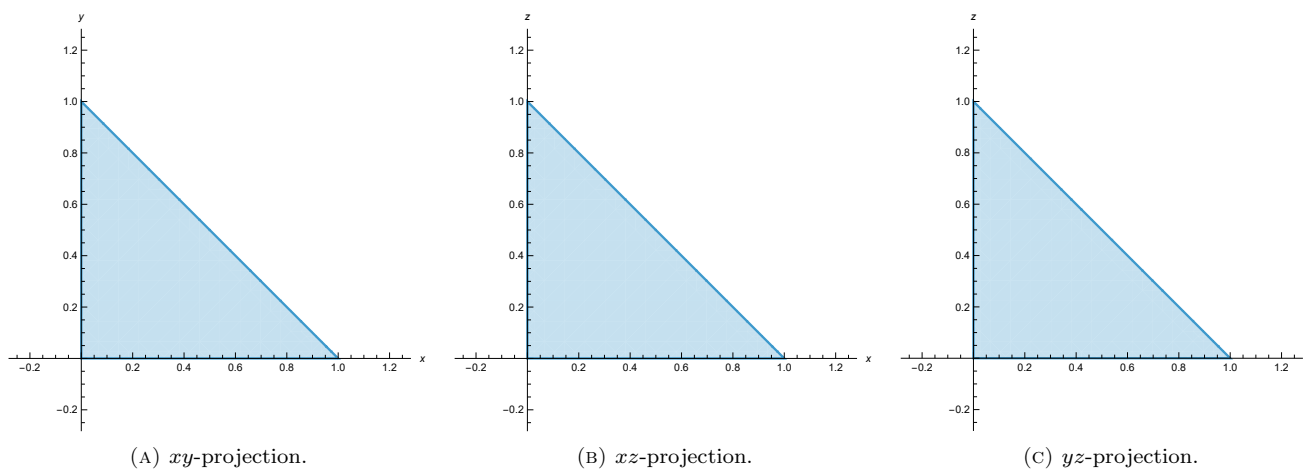


FIGURE 2. Projections of R_4 into the three coordinate planes.

The following triple integrals represent integrating over the region using the xy -projection:

$$\begin{aligned} \int_0^1 \int_0^{1-y} \int_0^{1-x-y} f(x, y, z) \, dz \, dx \, dy \\ \int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z) \, dz \, dy \, dx \end{aligned}$$

The following triple integrals represent integrating over the region using the xz -projection:

$$\begin{aligned} \int_0^1 \int_0^{1-z} \int_0^{1-x-z} f(x, y, z) \, dy \, dx \, dz \\ \int_0^1 \int_0^{1-x} \int_0^{1-x-z} f(x, y, z) \, dy \, dz \, dx \end{aligned}$$

The following triple integrals represent integrating over the region using the yz -projection:

$$\begin{aligned} \int_0^1 \int_0^{1-z} \int_0^{1-y-z} f(x, y, z) \, dx \, dy \, dz \\ \int_0^1 \int_0^{1-y} \int_0^{1-y-z} f(x, y, z) \, dx \, dz \, dy \end{aligned}$$

All of these triple integrals evaluate to $\frac{1}{6}$ when $f(x, y, z) = 1$.